복합 상관관계 전력 분석(CCPA) 고급 암호화 표준(AES) 모듈 대비

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Composite Correlation Power Analysis (CCPA) against Advanced Encryption Standard (AES) Modules

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Abstract

This paper presents a novel Composite Correlation Power Analysis (CCPA) method for cryptographic attacks, which significantly improves efficiency compared to traditional Correlation Power Analysis (CPA). Simulations on the hardware implementations of AES demonstrate that CCPA reduces the number of power traces required for key recovery by up to three times, enhancing cryptographic analysis accuracy and performance while lowering computational costs.

I. Introduction

Statistical power analysis against cryptographic devices has traditionally relied on three major techniques: Simple Power Analysis (SPA), Differential Power Analysis (DPA), and Correlation Power Analysis (CPA). Although SPA is relatively simple to perform as it only requires very few power traces, it shows critical limitations due to unrealistic assumptions about the target model [4]. DPA and CPA, on the other hand, rely on more complex statistical methods to identify patterns and correlations in large datasets [1]. However, despite being the most efficient among the previously mentioned techniques, CPA still needs a considerable number of power traces to extract the correct information, which often ends up being very costly [2].

This paper introduces a novel Composite Correlation Power Analysis (CCPA) method to recover the secret keyword from Advanced Encryption Systems. The CCPA technique exploits the non-linear encryption process at the S-box. Unlike the linear Shift Row and Mix Column stages of AES encryption, it is possible to extract distinct information from the input and output of the S-box, thereby reducing the number of power traces required to recover the secret key. CCPA demonstrates significantly improved efficiency in terms of the number of necessary power traces.

II. Proposed Model

2.1 Correlation Power Analysis (CPA)

Correlation Power Analysis (CPA) utilizes the data dependency of a cryptographic device by examining

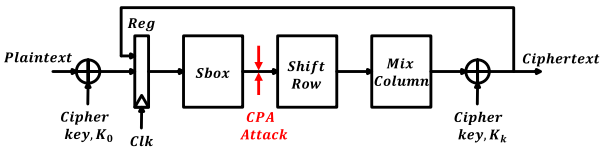


Figure 1. Block diagram of standard AES encryption (CPA on the S-box output)

a large set of power traces. It involves creating a hypothetical power model that assumes a correlation with the power consumption of the device [1]. The commonly used power models in CPA are based on the Hamming Distance (HD) and Hamming Weight (HW).

The HD model suggests that the power consumption of the device depends on the number of bit transitions when data changes from one state to another. It measures the difference between two states R and D and is represented mathematically as:

$$W_{HD} = aH(D \oplus R) + b \quad (1)$$

where a is a scalar gain between the Hamming distance and W_{HD} the power consumed, while the term b includes offsets, time-dependent components, and noise [1].

The HW model is similar but assumes that one of the states is zero. It considers the number of bit set to $'1'$ in the processed data, expressed as:

$$W_{HW} = aH(D \oplus 0) + b \quad (2)$$

After developing the power model, CPA involves analyzing the linear correlation between the hypothetical power model and the actual power consumption data from the cryptographic device. The correlation coefficient for a key guess k at time j is calculated using:

$$\Delta_k[j] = \frac{\sum_{i=1}^{N}(T_{i,k}-\overline{T_k})(H_{i,k}-\overline{H_k})}{\sqrt{\sum_{i=1}^{N}(T_{i,k}-\overline{T_k})^2}\sqrt{\sum_{i=1}^{N}(H_{i,k}-\overline{H_k})^2}} \quad (3)$$

where the summations are taken over the N samples $(i = 1, N)$ at each time step within the power traces $T_i(j), (j = 0, M-1)$ [5]. This correlation analysis helps to identify the correct cryptographic key. In AES modules, CPA attack is usually performed at the output of the S-box (Substitution Box) as shown in Figure 1. Hence, $H_{i,k}$ and $\overline{H_k}$ in (3) can be further written as:

$$H_{i,k} = HW_i\left[Sbox\left(C_i \oplus K_{k,l}\right)\right] \quad (4)$$
$$\overline{H_k} = \overline{HW[Sbox(C_i \oplus K_{k,l})]}$$

where C_i is the ith plain text, $K_{k,l}$ is the lth byte of the key guess k.

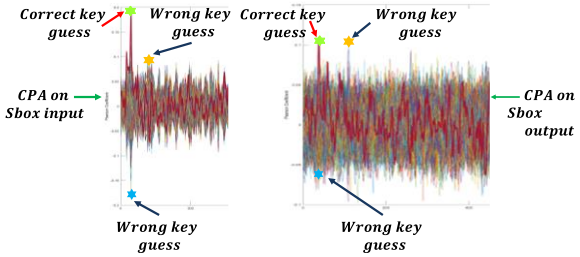


Figure 2. Correlation coefficients by key correct and incorrect guesses at two different stages of encryption

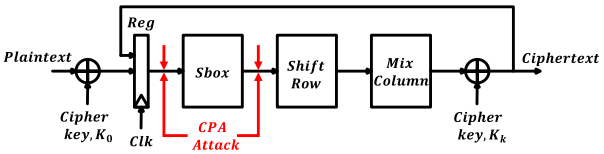


Figure 3. Block diagram of standard AES encryption (CPA on the S-box input and output)

2.2 The Proposed Composite Correlation Power Analysis (CCPA)

The main idea behind CCPA is that when CPA is measured at only one point, it is likely that no matter where this measurement was taken, there will be false positives caused by incorrect key guesses. However, an incorrect key guess is less likely to give two or more false positives at different states of encryption, while the correct key should give positive match at all the stages of encryption (Figure 2). This method is only applicable when the extracted sets of information at those two points are different. Because only the S-box performs non-linear encryption, its input and output reveal independent sets of information about the secret key. If CCPA is applied to the Shift Row or Mix Column stages, it would result in two linearly dependent sets of information.

Hence, applying the CPA technique at both the input and output of the S-box (Figure 3) and finding the combination of the correlation coefficients with the highest overall value help to extract more information from one power trace. This, consequently, reduces the overall number of power traces required to recover the cipher key.

Using (3) and (4), the correlation coefficients at the input and output can be expressed as:

$$\Delta_{Out,k}[j] = \frac{\sum_{i=1}^{N}(T_{i,k}-\overline{T_k})(H_{i,k}-\overline{H_k})}{\sqrt{\sum_{i=1}^{N}(T_{i,k}-\overline{T_k})^2}\sqrt{\sum_{i=1}^{N}(H_{i,k}-\overline{H_k})^2}} \quad (5)$$

$$\Delta_{In,k}[j] = \frac{\sum_{i=1}^{N}(T_{i,k}-\overline{T_k})(HW_i(C_i\oplus K_{k,l})-\overline{HW(C_i\oplus K_{k,l})})}{\sqrt{\sum_{i=1}^{N}(T_{i,k}-\overline{T_k})^2}\sqrt{\sum_{i=1}^{N}(HW_i(C_i\oplus K_{k,l})-\overline{HW(C_i\oplus K_{k,l})})^2}} \quad (6)$$

By defining α_k and β_k to be the maximum correlation coefficients for the input and output of the S-box over the M datapoints, two most reliable points for the key guess k can be found.

$$\alpha_k = \max_{j\in(0,M-1)}\{\Delta_{Out,k}[j]\}, \quad \beta_k = \max_{j\in(0,M-1)}\{\Delta_{In,k}[j]\} \quad (7)$$

Finally, the correct key guess is the key guess k that makes the multiplication of $\alpha_k * \beta_k$ maximum

$$\textbf{Correct key guess} = k = \max_{k\in\mathbf{K}}\{|\alpha_k\beta_k|\} \quad (8)$$

Ⅲ. Simulation results

The simulations were conducted using the power traces provided in [6], employing Electronic Codebook (ECB) [3] and masked mode implementations of AES on STM32F4 Microcontrollers. The dataset for ECB mode comprised 50,000 power traces, each consisting of 1,250 sample points, while the masked mode dataset contained 50,000 traces, each with 1,500 sample points. In both scenarios, cipher keys of 16 bytes were employed.

The results of CPA and CCPA attacks with 8,000 traces on Byte-0 of the AES cipher key in ECB mode are depicted in Figure 4 and Figure 5.

The same attacks with 10,000 traces on the Byte-0 of the AES cipher key in the masked mode are depicted in Figure 6 and Figure 7.

In the CPA plots, the dark red line represents the power trace of the correct key guess, while in the CCPA plots, the correct key guess has a distinctively high correlation coefficient. The data presents results for 256 different byte estimations. Figures 4-5 and 6-7 illustrate that CCPA exhibits a more pronounced differentiation between the correct key guess byte (Key byte #202) and the incorrect key guess bytes.

In line with this reasoning, a comparative analysis was conducted to generate Table 1, which outlines the minimum number of required traces for recovering the correct key byte in both CPA and CCPA for all 16 bytes of the AES key in ECB mode.

The data demonstrates that Byte-2 of the cipher key demanded a significantly higher number of traces

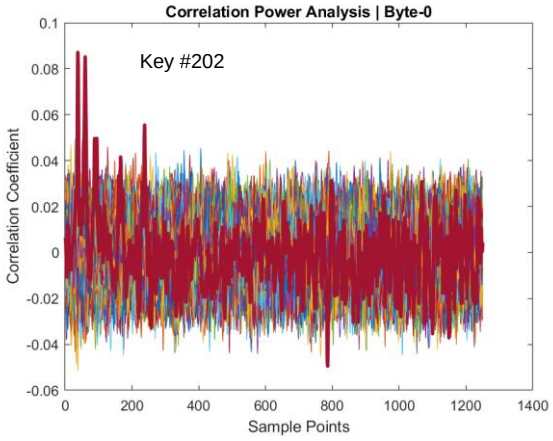


Figure 4. CPA attack on the Byte-0 of AES cipher key in ECB mode

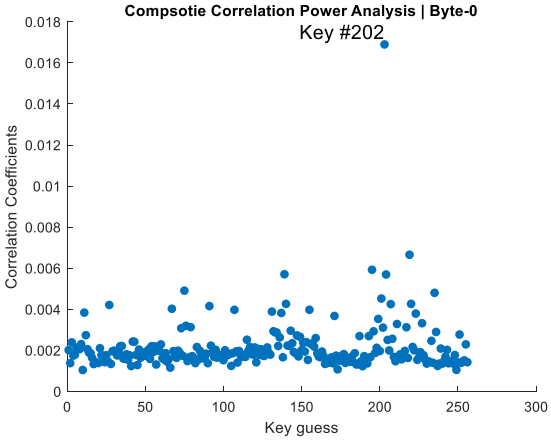


Figure 5. CCPA attack on the Byte-0 of AES cipher key in ECB mode

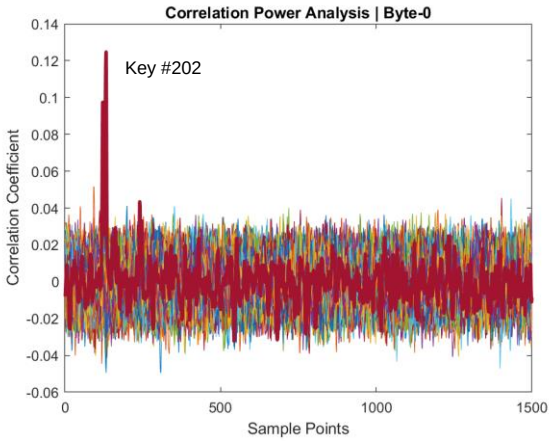


Figure 6. CPA attack on the Byte-0 of AES cipher key in the masked mode

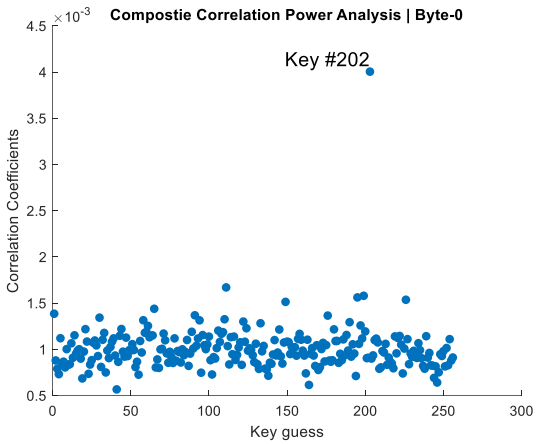


Figure 7. CCPA attack on the Byte-0 of AES cipher key in the masked mode

Table 1. Comparison of the minimum number of traces required for CPA and CCPA to recover the individual bytes of the cipher key

Bytes attacked	CPA on S-box input	CPA on S-box output	CCPA
Byte-0	≤ 5,000	≤ 2,000	1,000
Byte-1	≤ 5,000	≤ 2,000	600
Byte-2	≤ 5,000	≤ 5,000	1,500
Byte-3	≤ 5,000	≤ 2,000	800
Byte-4	≤ 5,000	≤ 1,500	500
Byte-5	≤ 5,000	≤ 300	300
Byte-6	≤ 5,000	≤ 2,000	1,100
Byte-7	≤ 5,000	≤ 2,000	900
Byte-8	≤ 5,000	≤ 1,200	500
Byte-9	≤ 5,000	≤ 3,000	1,000
Byte-10	≤ 5,000	≤ 3,000	800
Byte-11	≤ 5,000	≤ 2,000	1,000
Byte-12	≤ 5,000	≤ 3,000	1,000
Byte-13	≤ 5,000	≤ 3,000	1,000
Byte-14	≤ 5,000	≤ 3,000	900
Byte-15	≤ 5,000	≤ 2,000	400

compared to other bytes – specifically, 5,000 traces (red box) or more for CPA and 1,500 traces (green box) for CCPA. CCPA utilized three times fewer traces on average for each byte, showcasing its efficiency and effectiveness. The proposed method also achieved much better performance in terms of the necessary power traces to obtain the secret key compared with other CPA-based attack algorithms as shown in Table 2.

Table 2. Comparison of the proposed method with other CPA-based algorithms in terms of the minimum required traces to recover the secret key

CPA Attacks	Type of Setup	Minimum number of required traces
[4]	**CW305**	**10,000**
[5]	**SASEBO**	**6,000**
[6]	**SASEBO**	**3,000**
CPA	**STM32F4**	**5,000**
Proposed CCPA	**STM32F4**	**1,500**

Ⅳ. Conclusion and Future Research

The application of Composite Correlation Power Analysis (CCPA) in this study shows a marked improvement in efficiency compared to traditional Correlation Power Analysis (CPA). It requires significantly fewer traces to extract the cryptographic key, indicating CCPA's potential for more effective side-channel attacks on implementations of encryption systems like AES. Future research could explore integrating this technique into Machine Learning models and automating the process. Through the development of advanced models for the target device and the application of CCPA, the required number of power traces for key extraction can be significantly reduced.

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